CSE 400: Final Year Research and Project

1. Course Outcome (CO) and aligned Program Outcome (PO) of CSE-400 are as follows:

Term	COs	CO Statements	POs	Level
Spring	CO1	Identify a real-life problem / contemporary Issues that can be translated to an engineering and/or computing solution through design, development and validation with a focus on their potential contributions to ongoing research and future innovation.	PO12	H
	CO2	Identify outcomes and functional requirements of the proposed solution considering software and/or hardware specification and standards.	PO2	H
	CO3	Identify sub-components of a complex problem, prepare a timeline and appropriate budget using the project management skill.	PO11	H
	CO4	Investigate and analyse engineering /computing system /subsystem with given specifications and requirements	PO4	H
	CO5	Design and build the proposed solution of the problem considering the requirements and evaluate the system.	PO3	H
Fall	CO1	Use modern analysis and design tools in the process of designing and validating of a system and subsystem	PO5	M
	CO2	Assess social impacts and responsibilities of the engineering design project.	PO6	M
	CO3	Evaluate the ethical and professional implications of the proposed solution.	PO8	H
	CO4	Identify and validate the impact of environmental considerations and the sustainability of a system/subsystem of a complete project.	PO7	L
	CO5	Function effectively as a team.	PO9	M
	CO6	Present design project results through written technical documents and oral presentations.	PO10	H

2. Knowledge and attitude Profile

	Attribute
WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences
WK2	Conceptually based mathematics, numerical analysis, data analysis, statistics and the formal aspects of computer and information science to support detailed analysis and modeling applicable to the discipline
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of

	knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as professional responsibility of an engineer to public safety and sustainable development
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes

3. Range of Complex Engineering Problem Solving

Attribute	**Complex Engineering Problems have characteristics WP1 and some or all of WP2 to WP7:**
Depth of knowledge required	WP1: Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamentals-based, first principles analytical approach
Range of conflicting requirements	WP2: Involve wide-ranging or conflicting technical, non-technical issues (such as ethical, sustainability, legal, political, economic, societal) and consideration of future requirements
Depth of analysis required	WP3: Have no obvious solution and require abstract thinking, creativity and originality in analysis to formulate suitable models
Familiarity of issues	WP4: Involve infrequently encountered issues or novel problems
Extent of applicable codes	WP5: Address problems not encompassed by standards and codes of practice for professional engineering
Extent of stakeholder involvement and conflicting requirements	WP6: Involve collaboration across engineering disciplines, other fields, and/or diverse groups of stakeholders with widely varying needs
Interdependence	WP7: Address high level problems including many components or sub-problems that may require a systems approach

4. Range of Complex Engineering Activities

Attribute	**Complex activities mean (engineering) activities or projects that have some or all of the following characteristics:**
Range of resources	EA1: Involve the use of diverse resources including people, data and

	information, natural, financial and physical resources and appropriate technologies including analytical and/or design software
Level of interactions	EA2: Require optimal resolution of interactions between wide-ranging and/or conflicting technical, non-technical, and engineering issues
Innovation	EA3: Involve creative use of engineering principles, innovative solutions for a conscious purpose, and research-based knowledge
Consequences to society and the environment	EA4: Have significant consequences in a range of contexts, characterized by difficulty of prediction and mitigation
Familiarity	EA5: Can extend beyond previous experiences by applying principles-based approaches

5. List of Program Outcomes (POs) or graduate attributes:

Program Outcome	**Description**
PO1	Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions of complex engineering problems.
PO2	Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development (WK1 to WK4).
PO3	Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations as required (WK5).
PO4	Conduct investigations of complex engineering problems using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8).
PO5	Create, select and apply and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering problems (WK2, WK6).
PO6	When solving complex engineering problems, analyze and evaluate sustainable development impacts to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7).
PO7	Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9).
PO8	Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings (WK9).
PO9	Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences.
PO10	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one’s own work, as a member and leader in a team and to manage projects and in multidisciplinary environments.
PO11	Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the

	broadest context of technological change (WK8).
PO12	Demonstrate knowledge and understanding of the competences necessary to transform opportunities and ideas into a new business.

6. United Nations Sustainable Development Goals (SDGs)

SDG No.	SDG Name	Short Description
1	No Poverty	End poverty in all its forms everywhere.
2	Zero Hunger	End hunger, achieve food security, improve nutrition, and promote sustainable agriculture.
3	Good Health and Well-being	Ensure healthy lives and promote well-being for all at all ages.
4	Quality Education	Ensure inclusive and equitable quality education and promote lifelong learning opportunities.
5	Gender Equality	Achieve gender equality and empower all women and girls.
6	Clean Water and Sanitation	Ensure availability and sustainable management of water and sanitation for all.
7	Affordable and Clean Energy	Ensure access to affordable, reliable, sustainable, and modern energy.
8	Decent Work and Economic Growth	Promote sustained, inclusive, and sustainable economic growth and decent work.
9	Industry, Innovation and Infrastructure	Build resilient infrastructure, promote sustainable industrialization, and foster innovation.
10	Reduced Inequalities	Reduce inequality within and among countries.
11	Sustainable Cities and Communities	Make cities and communities inclusive, safe, resilient, and sustainable.
12	Responsible Consumption and Production	Ensure sustainable consumption and production patterns.
13	Climate Action	Take urgent action to combat climate change and its impacts.
14	Life Below Water	Conserve and sustainably use oceans, seas, and marine resources.
15	Life on Land	Protect, restore, and sustainably use terrestrial ecosystems and biodiversity.
16	Peace, Justice and Strong Institutions	Promote peaceful societies, justice, and accountable institutions.
17	Partnerships for the Goals	Strengthen global partnerships for sustainable development.

For CSE Graduates more relevant SDGs are:

- AI-based healthcare systems → SDG 3
- Smart education platforms → SDG 4
- Cyber security and digital safety → SDG 16
- Smart city or IoT systems → SDG 11
- Green computing or energy-efficient systems → SDG 7 and SDG 13
- Assistive technologies for accessibility → SDG 10

7. Guideline for Thesis Defense Presentation:

Thesis/Project presentation must include the following and may add additional as required by the research work:

- Introduction
- Related works
- Problem statements
- Thesis objectives
- Considered sustainability, societal and ethical facts
- Methodology
- Components of the System
- Developing the System
- Evaluating the System
- Conclusions
- Q & A

8. Guideline for Final Thesis Book:

Thesis book must include the followings two as Annexure A and B including other Annexures and relevant chapters as suggested by supervisors.

Annexure A: Engineering Considerations, Challenges and Remedies

This Annex should include the followings:

- Design Constraints
  - Ethical Considerations
  - Environmental Considerations
  - Social Constraints,
  - Sustainability on Environmental and Societal Context
- Relevant Challenges addressed
  - Addressed Knowledge Profile Attributes
  - Addressed Complex Engineering Problems
  - Addressed Complex Engineering Activities

Annexure B: Thesis Self-Assessment Report

(Template is given below)

9. Thesis Self-Assessment Report Template:

- Title of the Research:
- Name of supervisor and co-supervisor:
- Synopsis/Abstract of the research:
- Mapping of attained COs (Sample is given as below)

Term	**COs**	**CO Statements**	**POs**	**Level**	**Relevant Chapter/ Section/ Article**
Spring	CO1	Identify a real-life problem /	PO12	H	1.3,

		contemporary Issues that can be translated to an engineering and/or computing solution through design, development and validation with a focus on their potential contributions to ongoing research and future innovation.			4.3.2, and 5.2.3
	…	…	…		
Fall	CO1	Use modern analysis and design tools in the process of designing and validating of a system and subsystem	PO5	M	5.2 and 6.2
	…	…	…		

- Assessment Questions and Answer:
  (While answering the questions keep in mind the POs, KPs, CEPs, CEAs statements as given above, this is the justification of the previous mapping table, each question is relevant to each CO sequentially)
  i. What real-world problem did you address, what research gap did you identify, and how does your solution contribute to future innovation or advancement in the field?
  ii. How did you analyze the problem requirements, identify constraints, and determine the functional specifications of your proposed solution?
  iii. How did you plan, schedule, allocate resources, and manage risks throughout the project to ensure successful completion?
  iv. What investigation and analysis methods did you use to validate your assumptions, and how did the results influence your solution?
  v. How did you design, implement, and evaluate your solution to ensure that it satisfies the identified requirements and constraints?
  vi. Which modern engineering or computing tools did you use, why were they selected, and how did they support the development and validation of your project?
  vii. What are the societal impacts of your project, and how does your solution address the needs and responsibilities of stakeholders?
  viii. What ethical and professional issues are associated with your project, and how did you address them during the design and implementation process?
  ix. How does your project consider environmental sustainability, and what measures were taken to minimize negative environmental impacts?
  x. What was your role in the team, and how did collaboration among team members contribute to achieving the project objectives?
  xi. How would you communicate your project's objectives, methodology, results, and significance to both technical and non-technical audiences?
- Supervisor’s signature and student's signature Panel